

Shorthand for Thought: Compressing LLM Reasoning via Entropy-Guided Supertokens

Zhenyu Zhao, Sander Land, Daniel M. Bikel, Waseem Alshikh
 Writer, Inc.
 {zhenyu,sander,dan.bikel,waseem}@writer.com

Abstract

Reasoning in Large Language Models incurs significant inference-time compute, yet the token-level information structure of reasoning traces remains underexplored. We observe that reasoning tokens split into two functional types: low-entropy *structural* tokens (recurring phrases that scaffold the reasoning process) and higher-entropy *organic* tokens (problem-specific content that drives toward a solution). This asymmetry motivates a simple, model-agnostic compression pipeline: apply cross-word BPE merges on a model’s own reasoning traces to derive *supertokens* that capture frequent structural patterns, then teach the model to adopt them via supervised fine-tuning. Across three model families and five mathematical reasoning benchmarks, our approach shortens reasoning traces by 8.1% on average with no statistically significant accuracy loss on any model–benchmark pair. Beyond compression, supertokens act as interpretable reasoning-move annotations (backtracking, verification, strategy shifts), exposing the model’s high-level strategy at a glance. Analyzing transitions between structural categories reveals systematic differences between correct and incorrect traces: correct traces show productive recovery (backtracking followed by strategy shifts and verification), while incorrect traces are dominated by confusion cycles (repeated hedging and unresolved contradictions). These diagnostic signals suggest applications in reward shaping and early stopping for RL-based reasoning training.

1 Introduction

Large Language Models (LLMs) have achieved strong performance gains through chain-of-thought (CoT) reasoning, generating extended thinking tokens before producing answers (Wei et al., 2022; OpenAI, 2024; DeepSeek-AI, 2025). CoT reasoning has shown consistent benefits across mathematics, coding, and agentic planning (Lewkowycz et al., 2022; Chen et al., 2021; Yao et al., 2023b).

However, reasoning traces incur substantial computational cost, and efforts to compress them have focused on replacing explicit reasoning with latent states or reducing length through distillation (Deng et al., 2024; Chen et al., 2024). Comparatively little attention has been given to the *content* of reasoning outputs: the linguistic structure and information density of the traces themselves.

In this paper, we take a first step toward modeling LLM reasoning in a more structured and principled manner. We propose that reasoning traces are not a homogeneous stream of tokens, but rather a composition of two functionally distinct token populations that we term **structural** and **organic** reasoning tokens:

- **Structural tokens:** recurring, formulaic phrases that scaffold reasoning, such as backtracking cues (“Wait, hold on”), verification phrases (“let us verify”), and strategy shifts (“Let me try”), which organize the reasoning flow but carry little problem-specific information.

- **Organic tokens:** problem-specific content that drives toward a solution, including mathematical expressions, variable bindings, intermediate results, and reasoning direction changes.

We formalize this distinction using information theory. By measuring per-token conditional entropy across a large corpus of CoT traces, we find that structural tokens concentrate in frequent multi-token patterns whose continuations are near-deterministic once the phrase begins, while organic tokens exhibit substantially higher entropy. Wang et al. (2025) independently observe that roughly 80% of CoT tokens are generated with high confidence; we draw a different conclusion from this phenomenon: the low-entropy structural tokens represent a compression opportunity whose information content is not commensurate with the cost of generating them one by one.

We exploit this by applying cross-word BPE merges (Liu et al., 2025) independently on each model’s reasoning traces to derive model-specific reasoning *supertokens* appended to the base vocabulary. These supertokens predominantly capture structural patterns whose continuations are significantly more predictable than the corpus average. We teach models to adopt them via supervised fine-tuning (SFT), updating only the embedding layer, LM head, and a small number of transformer layers. The pipeline is model-agnostic: (1) collect reasoning traces, (2) derive supertokens via BPE merges, (3) fine-tune to adopt them.

Our approach yields reasoning traces that are 8.1% shorter on average across five benchmarks while preserving accuracy; all accuracy changes fall within the 95% confidence interval of zero (Appendix J). Unlike content-level compression methods that prune or latently reason steps, our vocabulary-level approach preserves the *complete, readable reasoning trace*: we compress the *expression* of reasoning (how structural phrases are tokenized) rather than the *content* (which steps are kept). Because the learned supertokens correspond to recognizable reasoning moves, the compressed traces expose the model’s high-level strategy as a readable sequence of structural signposts (Figure 3).

Our contributions are:

1. A **simple, model-agnostic pipeline** for reasoning compression requiring only BPE-based supertoken derivation and supervised fine-tuning.
2. An **information-theoretic analysis** of reasoning traces via the structural/organic decomposition, showing that supertokens predominantly capture low-entropy structural patterns.
3. **8.1% reasoning compression with no statistically significant accuracy loss** across three model families and five benchmarks, with evidence that supertokens double as **interpretable reasoning-structure annotations** exposing qualitative differences between correct and incorrect reasoning.

2 Related Work and Preliminaries

2.1 Related Work

Superword tokenization. Standard BPE tokenizers (Sennrich et al., 2016; Radford et al., 2019) enforce pretokenization boundaries that prevent tokens from spanning whitespace, limiting the efficiency gains available at larger vocabulary sizes. SuperBPE (Liu et al., 2025) addresses this by introducing a pretokenization curriculum that first learns subwords and then superwords bridging whitespace, achieving up to 33% fewer tokens than BPE at a fixed vocabulary size on general text. BoundlessBPE (Schmidt et al., 2025) independently proposes relaxing the pretokenization constraint to merge complete pretokens into superwords, similarly motivated by the observation that standard BPE yields diminishing returns at larger vocabulary sizes. Both works target pretraining efficiency on general text corpora. We apply the same core idea to a narrower domain: reasoning traces produced by a specific model family, where the distribution of high-frequency n-grams is structurally distinct from general text and amenable to targeted filtering.

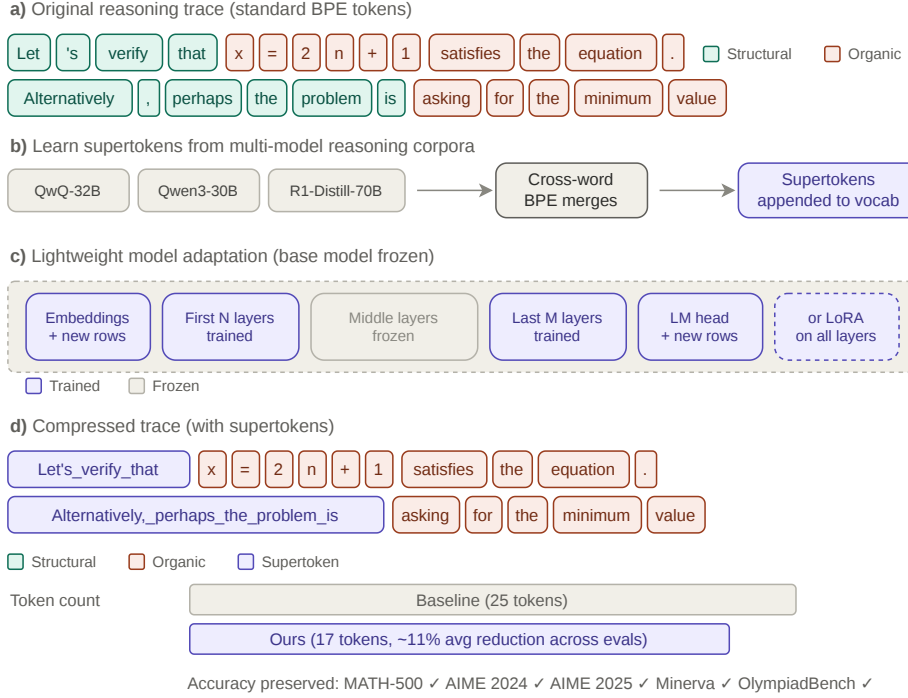


Figure 1: Overview of the pipeline: (1) derive reasoning-specific supertokens via BPE merges on reasoning traces, then (2) adapt the model to utilize these supertokens via supervised fine-tuning to produce shorter reasoning trajectories while preserving accuracy.

Vocabulary expansion without retraining. Kaplan et al. (2025) show that LLMs internally assemble multi-token word representations into coherent single vectors by intermediate layers, enabling vocabulary expansion without retraining core parameters. We follow this approach, initializing new supertoken embeddings by averaging their constituent token embeddings.

Structured reasoning. Several works impose meta-level structure on the reasoning process to improve efficiency and controllability (Yao et al., 2023a; Besta et al., 2024; Yao et al., 2024). PLAN-AND-BUDGET (Lin et al., 2026) decomposes tasks into structured sub-questions with complexity-based budget allocation, while Tan et al. (2025) incorporates symbolic logic scaffolds. These approaches modify the reasoning *process*; we instead analyze and compress its *output* at the vocabulary level.

Latent compression of reasoning traces. A separate line of work replaces discrete tokens with compressed latent representations. Token Assorted (Su et al., 2025) replaces initial reasoning steps with latent discrete tokens from a VQ-VAE, while Coconut (Hao et al., 2025) reasons entirely in continuous latent space. These approaches achieve strong compression but require specialized training objectives and lose interpretability.

Explicit CoT compression. A growing body of work directly compresses chain-of-thought traces at the content level. TokenSkip (Xia et al., 2025) prunes tokens by importance scores, achieving 30–40% reduction with <4% accuracy loss. R1-Compress (Yi et al., 2025) uses chunk-level compression with greedy search, reducing tokens by ~20% on MATH-500 with 0.6% accuracy drop. ConMax (Sun et al., 2025) formulates compression as RL-based reward optimization, yielding 43% reduction with 0.7% accuracy decrease. CtrlCoT (Chen et al., 2025) combines semantic abstraction with token pruning for 31% reduction on MATH-500. Extra-CoT (Huang et al., 2025) pushes extreme ratios, reporting >73% reduction on small models. These methods all operate at the content level, modifying or removing reasoning

Method	Approach	Token ↓	ΔAcc	Interpret.
TokenSkip	Step pruning	30–40%	<4% drop	✗
R1-Compress	Chunk compression	~20%	0.6% drop	✗
ConMax	RL compression	43%	0.7% drop	✗
CtrlCoT	Dual-granularity	31%	+7.6 pp*	✗
Extra-CoT	Extreme ratio	>73%	+0.6 pp*	✗
Coconut	Latent space	High	Varies	✗
Token Assorted	VQ-VAE latent	High	Varies	✗
Ours	Vocabulary	8.1%	~0 pp	✓

Table 1: Comparison with CoT compression methods. Content-level and latent methods achieve higher compression but alter or remove reasoning steps. Our vocabulary-level approach preserves the complete readable trace and is orthogonal to content-level methods (the two can be composed). *Relative to respective baselines on MATH-500.

steps from the trace. Table 1 positions our vocabulary-level approach relative to these methods and the latent compression work above.

3 Information-Theoretic Motivation

We now formalize the compression opportunity suggested by the structural/organic distinction, drawing on Shannon’s information-theoretic framework (Shannon, 1948; 1951). The key observation is that structural tokens concentrate in frequent *multi-token patterns* whose continuation tokens are highly predictable once the phrase begins: once a model commits to “*Let me reconsider*”, the remaining tokens of the phrase carry almost no information. We quantify this sequential redundancy via conditional entropy. Given a reasoning trace $T = (t_1, \dots, t_n)$ generated by a model with parameters θ , the conditional entropy at position i is:

$$H_\theta(t_i | t_{<i}) = - \sum_{v \in \mathcal{V}} P_\theta(t_i = v | t_{<i}) \log P_\theta(t_i = v | t_{<i}) \quad (1)$$

Let $\mathcal{M} \subset \{1, \dots, n\}$ denote positions within a multi-token merge span, with mean entropy $h_\mathcal{M} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} H_\theta(t_i | t_{<i})$. If a fraction $\rho = |\mathcal{M}|/n$ of tokens participate in merges, the theoretical compression ceiling is:

$$\Delta = \rho \cdot \left(1 - \frac{h_\mathcal{M}}{\log |\mathcal{V}|} \right) \quad (2)$$

When $h_\mathcal{M} \ll \log |\mathcal{V}|$, this simplifies to $\Delta \approx \rho$: nearly all tokens within merge spans can be absorbed, so a tokenization algorithm learning frequent patterns over reasoning traces should preferentially discover structural sequences.

4 Learning supertokens from reasoning traces

4.1 Vocabulary construction via SuperBPE

We apply SuperBPE (Liu et al., 2025) to a corpus of reasoning traces to derive a small set of *reasoning supertokens* that are appended to the base vocabulary of a pretrained LLM. The key design choices are (1) which token n-grams to consider as supertoken candidates, and (2) how to initialize the new token embeddings without retraining the transformer body.

Corpus and n-gram extraction. We generate reasoning traces from each target model on OpenThoughts3 (Guha et al., 2025) prompts; each model family receives its own supertokens from its own traces. The entropy analysis in Section 4.2 uses QwQ-32B (Qwen Team, 2025) as the representative model; cross-model generality is validated in Appendix H. N-gram counts are computed over reasoning traces only, with per-sample counts capped at 10 to prevent high-repetition samples from dominating merge selection (Land & Bartolo, 2024).

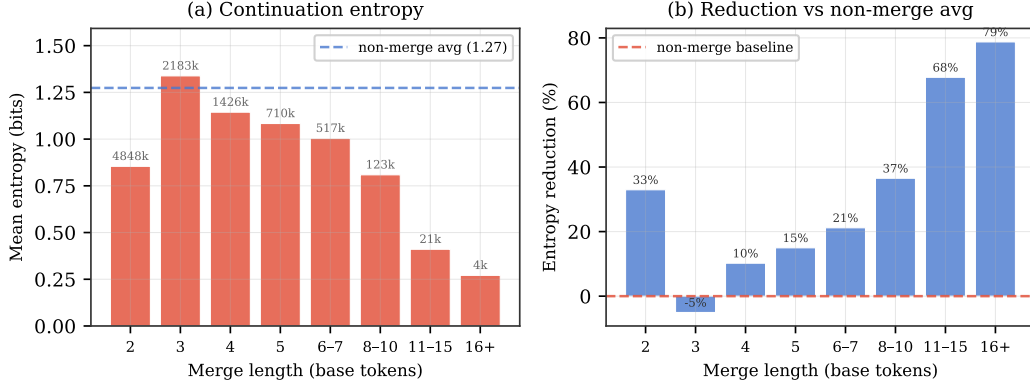


Figure 2: Continuation-token entropy by supertoken merge length. (a) Mean conditional entropy per bin (dashed line: non-merge average, 1.27 bits). The gap widens from 33% at length 2 to 79% at length 16+; length-3 merges are an outlier (−5%). (b) Relative entropy reduction ($1 - \bar{H}_{\text{cont}}/\bar{H}_{\text{non-merge}}$); longer merges capture progressively more deterministic continuations.

Filter design. Unfiltered BPE on reasoning text produces merges spanning syntactic boundaries (punctuation-initial sequences, cross-clause concatenations, multi-digit runs) unlikely to generalize. We define a *structural filter* restricting merges to four surface patterns: capitalized phrase-initial spans (e.g. “The answer is”), punctuation-plus-newline continuations, comma-led continuations, and space-prefixed single digits (multi-digit concatenations excluded). This reduces raw compression from ~28% to ~10% at 1000 merges (Figure 6, Appendix), producing semantically coherent reasoning phrases (Table 5, Appendix).

Merge budget. After filtering, we retain the top- $K=250$ merges ranked by corpus frequency, each introducing one new vocabulary entry (e.g., 151,669 $\rightarrow$ 151,919 for the Qwen-family models). Table 5 and Figure 6 (Appendix) show representative statistics.

Embedding initialization. New token embeddings are initialized by averaging the embeddings of their constituent tokens, providing a stable starting point that requires no additional forward passes.

Tokenizer application. At both training and inference time, the base tokenizer is applied first, and supertoken merges are applied as a post-processing step over the resulting token sequence. This keeps the extended tokenizer compatible with the original model and avoids modifying tokenizer internals.

4.2 Entropy Validation of Supertokens

We now test the prediction from Section 3: do the merges discovered by our supertokenization algorithm capture sequential redundancy in reasoning traces? We compute per-token conditional entropy across 10,000 traces (115M tokens) from OpenThoughts-1.2M using QwQ-32B as both generator and scorer, align each base token with the supertoken-based tokenization via character offsets, and classify tokens into three roles: **non-merged**, **first** (the opening token of a merge span), and **continuation** (subsequent tokens completing the phrase).

Merge length scaling. Our supertoken merges cover $\rho = 15.2\%$ of all token positions. As shown in Figure 2, continuation-token entropy decreases with merge length: length-2 merges already show a 33% entropy reduction relative to non-merge tokens, and merges of length ≥ 16 reach 79% reduction. This confirms that the merges target *sequential* redundancy, i.e., phrases where continuations become near-deterministic once the opening token is observed (see also Figure 7, Appendix). Length-3 merges are an exception, displaying

Benchmark	Baseline		SFT			Improvement	
	Acc (%)	Avg. Tok. ($\downarrow$)	Acc (%)	Avg. Tok. ($\downarrow$)	SuperBPE (%)	Δ Acc (%)	Δ Avg. Tok. (%) ($\downarrow$)
QWQ-32B							
AIME'24	77.50	14082	77.7	13160	7.6	+0.2	-6.5
AIME'25	69.20	16651	72.7	14880	7.7	+5.5	-10.6*
MATH-500	80.60	4441	80.5	4183	10.7	-0.1	-5.8*
Minerva	34.0	5324	33.5	5418	8.15	-1.5	-1.1
Olympiad	55.8	9584	56.5	9256	11.7	+1.3	-3.0*
Average	63.4	9984.8	64.2	9379	9.17	+1.3	-6.4*
QWEN-3-30B-A3B							
AIME'24	91.3	16305	90.3	15246	10.7	-1.2	-6.5
AIME'25	84.0	19491	84.1	18517	11.1	+0.1	-5.0
MATH-500	85.7	5488	86.8	4849	12.9	+1.2	-11.6*
Minerva	37.4	3595	37.0	3571	9.7	-1.1	-0.66
Olympiad	65.3	13309	66.8	12372	11.7	+2.3	-7.0*
Average	74.7	11638	73	10910	11.2	-1.3	-6.3*
DeepSeek-R1-Llama-70B-Distill							
AIME'24	66.7	9059	70.2	7519	12.1	+5.1	-17.0*
AIME'25	54.5	11332	53.4	9390	11.6	-2.0	-17.0*
MATH-500	78.1	2701	74.8	2541	12.2	-3.1	-5.9*
Minerva	31.6	3239	31.4	3126	9.54	-0.6	-3.5*
Olympiad	51.0	6373	49.5	6370	11.4	-2.9	-0.1
Average	56.4	6541	55.9	5749	11.4	-0.8	-11.5*

Table 2: SFT encourages supertoken adoption, reducing reasoning trace length while preserving accuracy across three model families and five math benchmarks. All accuracy deltas fall within the 95% confidence interval of zero; * marks token reductions whose 95% CI excludes zero ($p < 0.05$). See Appendix J for methodology. Table 9.

entropy on par with non-merge tokens, likely because they contain a high concentration of transitioning tokens (Wang et al., 2025).

5 Experiments and Results

5.1 Experimental setup

Supertoken adoption training. We fine-tune each base model on OpenThoughts3 (Guha et al., 2025) with the extended tokenizer applied to assistant turns only. We compare three SFT modes: (i) *embedding-only* (new supertoken rows in embed_tokens and lm_head); (ii) *LoRA* (Hu et al., 2022) ($r=16$, $\alpha=32$ on attention projections plus embedding rows); and (iii) *layer-unfreezing* (first N and last M transformer layers plus embeddings). In all modes, gradient hooks zero out base-vocabulary rows so only supertoken embeddings receive updates.

Evaluation. We evaluate on five mathematical reasoning benchmarks (AIME'24, AIME'25, MATH-500 (Hendrycks et al., 2021), Minerva, OlympiadBench (He et al., 2024)) in a zero-shot setting using vLLM (Kwon et al., 2023) on 2 H200 GPUs, running each evaluation 5 times and reporting mean accuracy, completion tokens, and supertoken adoption rate (fraction of reasoning tokens from the expanded vocabulary). Wall-clock latency reductions generally track or exceed token reductions, as shown in in Appendix G.

5.2 Adoption of supertokens and compression of reasoning trajectories

As shown in Table 2, SFT on a retokenized dataset consistently drives supertoken adoption across all three model families and five benchmarks, reducing reasoning trace length by 6–12% on average while preserving accuracy. The two Qwen-family models (QwQ-32B (Qwen

Category	Description	Representative Supertokens
■ Backtracking	Self-correction; pausing to reconsider	Wait, hold on But wait Wait, no
■ Hedging	Expressing uncertainty or tentativeness	, maybe But maybe , so maybe
■ Verification	Checking or validating intermediate results	, right , correct let's check
■ Counterargument	Introducing a contradiction or counterpoint	, but , but the , but we But the
■ Strategy Shift	Changing approach or trying a new method	Let's Let me Now, let's Let's try
■ Problem Ref.	Re-reading or referencing the problem	, the problem says But the problem
■ Sequencing	Organizing steps or structural ordering	First, Similarly, Given that Now, the
■ Reasoning	Forward-flowing connectives	, so , and , which is Therefore,
■ Computation	Digits or single-letter variables	1 2 , x , n , k

Table 3: Supertoken structural taxonomy. Each merge in the SuperBPE vocabulary is classified into one of nine structural categories based on its decoded text. Representative supertokens are shown for each category.

Team, 2025), Qwen3-30B-A3B (Yang et al., 2025)) show robust accuracy (≤ 1.3 pp average change); DeepSeek-R1-Distill-Llama-70B (DeepSeek-AI, 2025) achieves the largest compression (up to -17% on AIME) but exhibits a -3.1 pp accuracy drop on MATH-500, which we attribute to its Llama-based tokenizer (Dubey et al., 2024) having a substantially different subword inventory that makes supertoken integration more disruptive. None of the 15 accuracy deltas are statistically significant (all fall within the 95% CI of zero; Appendix J), while token reductions marked * in Table 2 are significant ($p < 0.05$), confirming that the compression gains are real.

Comparing training modes (Table 7), we find that LoRA achieves supertoken adoption but does not yield compression, whereas partial layer unfreezing produces both. This suggests that learning to *use* supertokens and learning to *compress with* them are distinct processes that require different degrees of model flexibility.

5.3 Supertokens as structural signposts in reasoning trajectories

Beyond compression, supertokens enable interpretable visualization of reasoning structure. We classify supertokens into the taxonomy shown in Table 5.2 using deterministic keyword-based rules applied in priority order to each merge’s decoded text (full rule specification in Appendix A.2), which identifies key reasoning moves and transforms each trace into a sequence of interpretable modular blocks.

Figure 3 shows two reasoning traces partitioned into reasoning blocks joined and demarcated by the structural supertokens, highlighting the logical flow, key transitions, and critical turns throughout each trajectory. The structural supertokens transform the highly dense and opaque token-level view into a meta-token-level one. The correct trace exhibits a balanced structure: the model explores strategies, verifies intermediate results (15% of structural tokens), and backtracks sparingly (7%). In contrast, the incorrect trace is dominated by repeated backtracking (31%) with almost no verification (3%); the model second-guesses itself repeatedly but never checks its work.

5.4 Supertokens provide insights into quality of reasoning traces

We performed aggregated analysis on the supertoken transitions across all evaluation samples’ reasoning trajectories and found that the *transitions* between structural categories

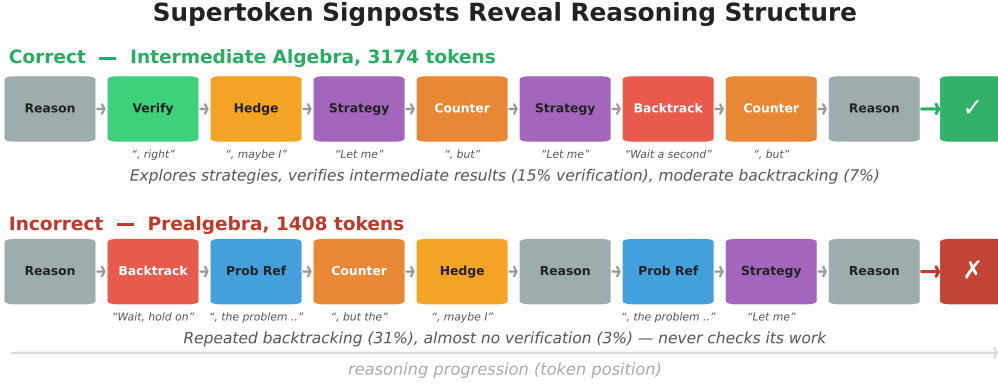


Figure 3: Supertoken signposts reveal reasoning structure. Each block represents a structural reasoning move detected via supertokens. The correct trace (top) shows diverse, balanced reasoning with verification; the incorrect trace (bottom) is dominated by repeated backtracking without verification.

From	To	Ratio	Interpretation
<i>Problematic transitions</i> (over-represented in incorrect traces)			
Sequencing	Sequencing	3.7×	Repeated reorganization without progress
Problem Ref.	Hedging	2.1×	Re-reads problem → becomes uncertain
Counterargument	Problem Ref.	2.0×	Contradiction → re-reads problem (confusion loop)
Hedging	Hedging	1.4×	Sustained unresolved uncertainty
Counterargument	Counterargument	1.3×	Contradictions accumulating unresolved
<i>Productive transitions</i> (over-represented in correct traces)			
Problem Ref.	Strategy Shift	3.0×	Re-reads problem → devises new plan
Verification	Strategy Shift	2.1×	Checks result → adapts approach
Reasoning	Verification	1.6×	Forward reasoning → verifies work
Backtracking	Strategy Shift	1.5×	Recognizes error → pivots constructively

Table 4: Diagnostic supertoken transitions. We compare the normalized transition rates between correct and incorrect reasoning traces. *Problematic* transitions are significantly over-represented in incorrect traces and indicate unproductive reasoning patterns; *productive* transitions are over-represented in correct traces and indicate healthy problem-solving behavior.

carry strong diagnostic signal about reasoning quality. Figure 8 (Appendix) shows the full transition probability matrix.

Diagnostic transitions and composite metrics. Splitting the transition matrix by answer correctness reveals two systematic patterns (Table 4). *Incorrect* traces are dominated by **confusion cycles**: problem-reference→hedging (2.1× over-represented), contradictions→re-reading (2.0×), and hedging→hedging (1.4×), indicating the model loops without making progress. *Correct* traces instead show **productive recovery**: re-reading leads to a new strategy (3.0×) and verification is followed by adaptation (2.1×).

We formalize these patterns into three composite metrics: (1) **productive recovery rate** (fraction of post-difficulty transitions leading to a strategy shift or verification: 23.3% correct vs. 17.4% incorrect, a 34% gap); (2) **confusion cycle rate** (unproductive bigrams like hedging→hedging: 50% higher in incorrect traces); and (3) **verification inflow rate** (transitions entering verification: 14% higher in correct traces). Together, these show that

reasoning quality depends not on which moves a model makes but on how it sequences them, providing actionable signals for reward shaping and early stopping (§6.1).

6 Discussion

6.1 Implications and future directions

Because supertokens are part of the model’s vocabulary and generated autoregressively, they can be monitored in real time during inference, serving as a lightweight control surface for reasoning. A system could detect confusion cycles (e.g., consecutive problem-reference→hedging transitions) and intervene by triggering early termination or injecting a verification prompt. This is particularly relevant for RL-based training methods such as GRPO (Shao et al., 2024): confusion cycle detection could provide an early stopping signal, abandoning unproductive trajectories before they consume thousands of tokens, while the productive recovery rate offers a natural reward shaping signal (Lightman et al., 2024).

More broadly, our findings suggest that reasoning traces are not free-form text but a distinct linguistic register with its own grammar of moves (backtrack, verify, pivot, hedge). Roughly 78% of supertokens fall into the *Reasoning* and *Computation* categories, while the remaining 22% constitute structural signposts. This motivates a two-tier view: an *explicit* structural layer governing reasoning flow and an *implicit* content layer performing computation, where the scaffolding could be explicitly planned or constrained while content is generated autoregressively.

6.2 Limitations

Compression–interpretability trade-off. Our approach achieves more modest compression (~8.1%) compared to content-level CoT compression methods that report 30–70% reductions (Table 1). This is a deliberate design choice: by operating at the vocabulary level, we compress the *tokenization* of reasoning rather than its *content*, preserving the complete human-readable trace. Our approach enables the structural analysis, visualization, and diagnostic transition metrics presented in Sections 5.3–5.4. Because the two strategies operate at different levels (vocabulary vs. content), they are orthogonal and could be composed for greater overall savings.

7 Conclusion

We presented a simple, model-agnostic pipeline for compressing LLM reasoning traces via entropy-guided supertokens. By applying cross-word BPE merges to reasoning corpora and fine-tuning only a thin parameter surface (embeddings, LM head, and a small number of transformer layers), we achieve 8.1% average compression across three model families and five benchmarks with no statistically significant accuracy loss.

Beyond compression, supertokens serve as interpretable structural annotations that expose the high-level reasoning strategy of a model at a glance. We introduced a nine-category taxonomy of structural reasoning moves and showed that the transitions between these categories carry strong diagnostic signal: correct traces are characterized by productive recovery patterns (re-reading leading to new strategies, verification followed by adaptation), while incorrect traces exhibit confusion cycles (repeated hedging, unresolved contradictions, backtracking without verification).

These findings open several directions for future work. The diagnostic transitions and composite metrics we defined (productive recovery rate, confusion cycle rate, verification inflow) could be integrated as reward signals or early stopping criteria in RL-based training pipelines such as GRPO, redirecting compute away from unproductive reasoning trajectories. More broadly, the structural/organic decomposition motivates a two-tier view of reasoning where the scaffolding layer is explicitly planned or constrained while the content layer is generated autoregressively, offering a path toward more efficient and controllable reasoning in LLMs.

References

- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and Torsten Hoefler. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 17682–17690, 2024.
- Kaiwen Chen et al. CtrlCoT: Dual-granularity chain-of-thought compression for controllable reasoning. *arXiv preprint arXiv:2601.20467*, 2025.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- Xin Chen, Hanxian Huang, Yanjun Gao, Yi Wang, Jishen Zhao, and Ke Ding. Learning to maximize mutual information for chain-of-thought distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 6857–6868, 2024.
- Tri Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. *arXiv preprint arXiv:2307.08691*, 2023.
- DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Yuntian Deng, Yejin Choi, and Stuart Shieber. From explicit CoT to implicit CoT: Learning to internalize CoT step by step. *arXiv preprint arXiv:2405.14838*, 2024.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aieleen Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Etash Guha, Ryan Marten, Sedrick Keh, Negin Raoof, Georgios Smyrnis, Hritik Bansal, Marianna Nezhurina, Jean Mercat, Trung Vu, Zayne Sprague, Ashima Suvarna, Benjamin Feuer, Liangyu Chen, Zaid Khan, Eric Frankel, Sachin Grover, Caroline Choi, Niklas Muennighoff, Shiye Su, Wanxia Zhao, John Yang, Shreyas Pimpalgaonkar, Kartik Sharma, Charlie Cheng-Jie Ji, Yichuan Deng, Sarah Pratt, Vivek Ramanujan, Jon Saad-Falcon, Jeffrey Li, Achal Dave, Alon Albalak, Kushal Arora, Blake Wulfe, Chinmay Hegde, Greg Durrett, Sewoong Oh, Mohit Bansal, Saadia Gabriel, Aditya Grover, Kai-Wei Chang, Vaishaal Shankar, Aaron Gokaslan, Mike A. Merrill, Tatsunori Hashimoto, Yejin Choi, Jenia Jitsev, Reinhard Heckel, Maheswaran Sathiamoorthy, Alexandros G. Dimakis, and Ludwig Schmidt. Openthoughts: Data recipes for reasoning models, 2025. URL <https://arxiv.org/abs/2506.04178>.
- Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuan-dong Tian. Training large language models to reason in a continuous latent space, 2025. URL <https://arxiv.org/abs/2412.06769>.
- Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. OlympiadBench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *Advances in Neural Information Processing Systems*, 34:7294–7305, 2021.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Jiayi Huang et al. Towards efficient large language reasoning models via extreme-ratio chain-of-thought compression. *arXiv preprint arXiv:2602.08324*, 2025.

- Guy Kaplan, Matanel Oren, Yuval Reif, and Roy Schwartz. From tokens to words: On the inner lexicon of llms, 2025. URL <https://arxiv.org/abs/2410.05864>.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, pp. 611–626, 2023.
- Sander Land and Max Bartolo. Fishing for magikarp: Automatically detecting under-trained tokens in large language models, 2024. URL <https://arxiv.org/abs/2405.05417>.
- Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. Solving quantitative reasoning problems with language models. *Advances in Neural Information Processing Systems*, 35:3843–3857, 2022.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations*, 2024.
- Junhong Lin, Xinyue Zeng, Jie Zhu, Song Wang, Julian Shun, Jun Wu, and Dawei Zhou. Plan and budget: Effective and efficient test-time scaling on reasoning large language models, 2026. URL <https://arxiv.org/abs/2505.16122>.
- Alisa Liu, Jonathan Hayase, Valentin Hofmann, Sewoong Oh, Noah A Smith, and Yejin Choi. SuperBPE: Space travel for language models. *arXiv preprint arXiv:2503.13423*, 2025.
- OpenAI. Learning to reason with LLMs. 2024. Blog post, September 2024.
- Qwen Team. QwQ-32b technical report. *arXiv preprint arXiv:2412.15115*, 2025.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI Blog*, 2019.
- Craig W. Schmidt, Varshini Reddy, Chris Tanner, and Yuval Pinter. Boundless byte pair encoding: Breaking the pre-tokenization barrier, 2025. URL <https://arxiv.org/abs/2504.00178>.
- Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pp. 1715–1725, 2016.
- Claude E Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.
- Claude E Shannon. Prediction and entropy of printed english. *The Bell System Technical Journal*, 30(1):50–64, 1951.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Li, Mingchuan Zhang, Y.K. Zhang, Yu Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Dijia Su, Hanlin Zhu, Yingchen Xu, Jiantao Jiao, Yuandong Tian, and Qingqing Zheng. Token assorted: Mixing latent and text tokens for improved language model reasoning, 2025. URL <https://arxiv.org/abs/2502.03275>.
- Shicheng Sun et al. ConMax: Confidence-maximizing compression for efficient chain-of-thought reasoning. *arXiv preprint arXiv:2601.04973*, 2025.
- Xingwei Tan, Peng Yu, Haoyuan He, Jiaze Chen, Xiaohui Hu, and Yuan-Fang Li. Enhancing logical reasoning in language models via symbolically-guided Monte Carlo process supervision, 2025. URL <https://arxiv.org/abs/2505.20415>.

Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for LLM reasoning. *arXiv preprint arXiv:2506.01939*, 2025.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.

Heming Xia, Yongqi Ge, Lun Chen, Qingxiu Zhang, Hao Li, et al. TokenSkip: Controllable chain-of-thought compression in LLMs. In *Proceedings of EMNLP*, 2025. arXiv:2502.12067.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Wang, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models, 2023a. URL <https://arxiv.org/abs/2305.10601>.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations*, 2023b.

Yao Yao, Zuchao Li, and Hai Zhao. GoT: Effective graph-of-thought reasoning in language models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 2901–2921, 2024.

Quanjin Yi, Xiao Wu, Hai Zhu, et al. R1-Compress: Long chain-of-thought compression via chunk compression and search. *arXiv preprint arXiv:2505.16838*, 2025.

Jeffrey Zhou et al. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

A Deeper Look into Reasoning Supertokens

Table 5 shows representative supertokens under different filtering rules. The *structural filter* (Section 4.1) removes high-frequency patterns unrelated to reasoning (e.g., “is the”, “in the”) and possessive artifacts starting with “s”, while retaining compositional reasoning phrases such as “Let’s check” and “Let’s reconsider”.

Token	No filter	Filtered
<i>Kept by both strategies</i>		
. \n	261,553	261,553
. \n\n	254,138	254,138
: \n	243,970	243,970
, _the	242,352	242,352
_ 1	264,450	264,450
, _then	193,242	193,242
What _if	136,224	136,224
So _the	135,171	135,171
<i>Removed by filter</i>		
_is _the	241,551	–
_is _a	241,413	–
_in _the	231,043	–
_of _the	230,200	–
's _check	219,883	–
's _assume	165,624	–
impl ies	196,004	–
? _No . \n	210,179	–
<i>Surfaced by filter</i>		
Let 's	–	189,651
Is _it	–	156,033
Let 's _check	–	152,862
Is _it _possible	–	151,487
This _is	–	147,188
Is _there	–	142,627
Wait , _if	–	127,788
Let 's _assume	–	124,009
This _implies	–	105,077
Let 's _reconsider	–	47,570
Let 's _consider	–	45,207

Frequencies at 150-merge budget. – indicates token absent from that merge table.

Table 5: Representative super-tokens from unfiltered and filtered BPE. The filter removes syntactically promiscuous merges and math artifacts, and by admitting apostrophes surfaces a compositional family of reasoning-specific phrases built on `Let 's` as a shared prefix.

A.1 Exemplary Reasoning Traces

Figures 4 and 5 show two complete reasoning traces rendered at the token level, with supertoken merge positions colored by their structural category (Table 5.2). Each figure consists of a compact *ribbon* overview (top) showing the category distribution across the full trace, and three *zoomed windows* that reveal token-level detail at structurally interesting points.

The correct trace (Figure 4) exhibits a diverse mix of categories throughout: strategy shifts (purple) and verification (green) appear regularly, interspersed with forward reasoning. In contrast, the incorrect trace (Figure 5) is dominated by backtracking (red) and counterargu-

ments (orange) with notably fewer verification events, consistent with the confusion-cycle pattern described in Section 5.4.



Figure 4: Correct reasoning trace with supertoken categories. Top: ribbon overview showing category distribution across all 3,174 tokens. Bottom: three zoomed windows at structurally interesting positions. Colors correspond to the taxonomy in Table 5.2; gray tokens are non-merged (organic content).

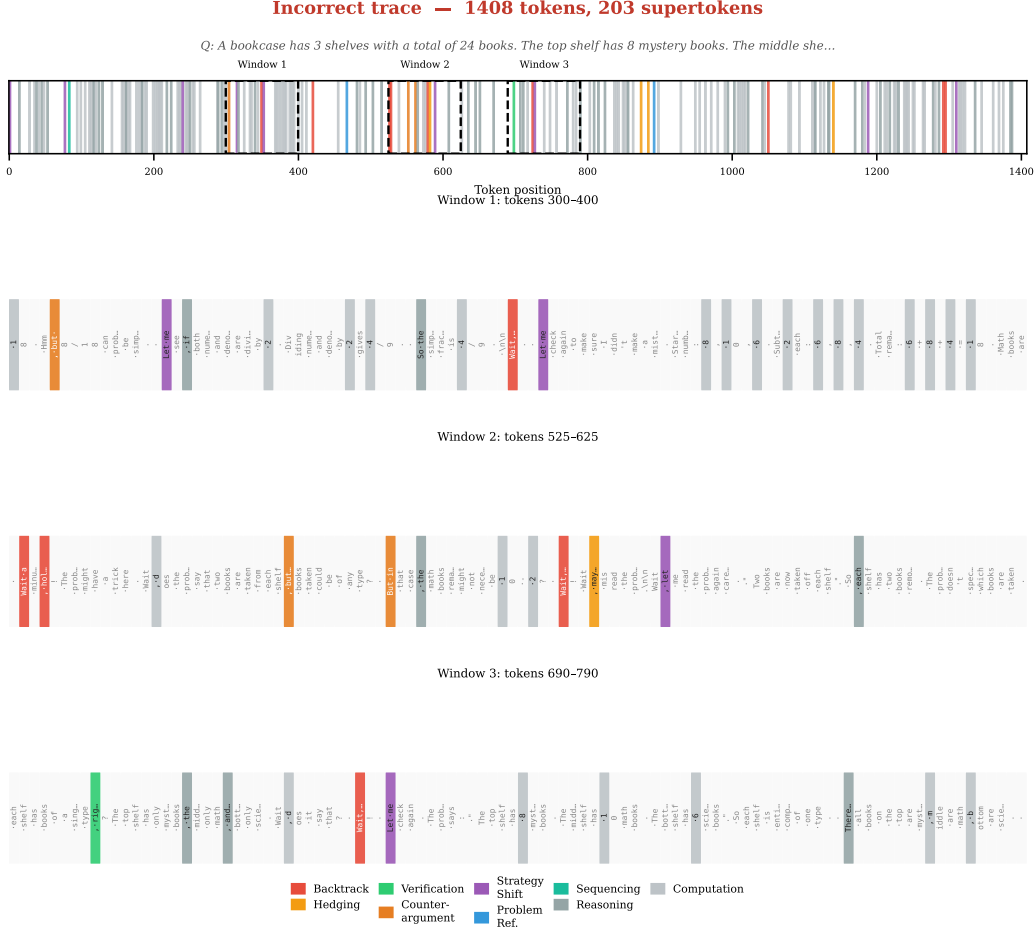


Figure 5: Incorrect reasoning trace with supertoken categories. The ribbon overview reveals a higher density of backtracking (red) and counterargument (orange) tokens compared to the correct trace. Zoomed windows show repeated self-correction cycles without verification, illustrating the confusion patterns identified in Section 5.4.

A.2 Taxonomy Construction Details

The nine-category taxonomy in Table 5.2 is applied via a deterministic, rule-based classifier that operates on the decoded text of each supertoken merge. Classification proceeds by evaluating pattern-matching rules in a fixed priority order; the first matching rule determines the category. No machine-learned component or manual per-token annotation is involved: the same script reproducibly assigns categories to any set of merges.

Rule summary. Table 6 lists the keyword or regex pattern that triggers each category, ordered by evaluation priority (highest first). Higher-priority rules ensure that composite phrases are resolved correctly; for example, “Wait, let’s check” matches *Backtracking* (via the “wait” prefix) before it could match *Verification* (via “check”) or *Strategy Shift* (via “let’s”).

Pri.	Category	Key patterns (decoded text)	N
1	Backtracking	starts with “Wait”; contains “hold on”, “but actually”	16
2	Hedging	contains “maybe”	12
3	Verification	contains “check”, “matches”, “which is what”; exact match: “correct”, “yes”, “right”, “good”, “impossible”	12
4	Problem Ref.	contains “problem” or “it says”	8
5	Strategy Shift	starts with “Let’s”, “Let me”, “So let”, “But let”	12
6	Sequencing	starts with “First”, “Similarly”, “Given that”, “Now, the”	5
7	Computation	single digit (0–9) or single-letter variable (excl. “a” and “I” to avoid articles/pronouns)	36
8	Counterargument	starts with “But ” or “, but”	25
9	Reasoning	comma-led connectives (“, so”, “, and”, “, which”, ...); “Therefore”; “I think”; fallback: any “, ” + word	124

Table 6: Taxonomy classification rules applied in priority order. N = number of the 250 supertokens assigned to each category. All 250 merges are classified; no merges fall into an “other” residual category.

Coverage and ambiguity. Of the 250 supertokens, all are classified by the deterministic rules (0 fall into the residual “other” category). Because rules are evaluated in priority order, composite phrases that could match multiple categories (e.g., “, but maybe the problem” contains keywords for Counterargument, Hedging, and Problem Reference) are resolved deterministically. In this example, the “maybe” rule (priority 2) fires before “problem” (priority 4), yielding *Hedging*, reflecting that the dominant pragmatic function of the phrase is expressing uncertainty rather than referencing the problem. We verified the priority ordering by manually inspecting all 25 supertokens that match keywords from two or more categories; in each case the highest-priority rule assigns the pragmatically dominant function.

The full classification script and its output (a JSON mapping from token ID to category) are released with our code, enabling exact reproduction of all taxonomy-dependent analyses in Sections 5.3–5.4.

B Additional Experiment Details

This section describes the compute environment, hyperparameters, and ablation studies used in our experiments.

B.1 Compute and infrastructure

All SFT runs use $8 \times \text{H200}$ GPUs. QwQ-32B and Qwen3-30B-A3B each train for 1 epoch on OpenThoughts3 ($\sim 114\text{K}$ samples); DeepSeek-R1-Distill-Llama-70B uses the same data and epoch count. Total wall-clock training time ranges from 4–12 hours depending on model size.

B.2 Training hyperparameters

Training uses a per-device batch size of 1 with 16 gradient accumulation steps (effective batch size 128), a learning rate of 2×10^{-4} with cosine decay and 3% linear warmup, for 1 epoch. We use bfloat16 mixed precision, Flash Attention 2 (Dao, 2023), gradient checkpointing, and Fully Sharded Data Parallel (FSDP) with full sharding. Checkpoints are saved every 100 steps. A separate learning rate can optionally be applied to the embedding/ lm_head parameters independently from the backbone learning rate.

B.3 Minimum training ablation

To understand the minimum training needed for supertoken adoption, we tested two reduced configurations on Qwen3-30B-A3B: (i) masking losses from all tokens except merged supertokens; and (ii) LoRA with ranks $r \in \{1, 2, 4, 8, 16\}$. Results for the LoRA rank sweep are presented in Section D.

C Embedding-only training vs. LoRA vs. Sub-layer SFT

Here we present the results of the three different training approaches we tested: Embedding-only, LoRA, and partial-layer unfreezing. All three methods freeze the base model’s pre-trained weights and differ only in which additional parameters receive gradient updates during SFT. Embedding-only trains just the new supertoken rows in `embed_tokens` and `lm.head`. LoRA adds low-rank adapters ($r=16$, $\alpha=32$) to attention projections. Partial unfreezing selectively unfreezes the first N and last M transformer layers alongside the embedding rows.

Method	Benchmark	Acc (%)	Avg. Tok. ($\downarrow$)	Supertoken (%)
Baseline (no SFT)	AIME’24	77.50	14082	—
	AIME’25	69.20	16651	—
	MATH-500	80.60	4283	—
Embedding-only	AIME’24	70	13899	7.2
	AIME’25	57.3	17342	7.9
	MATH-500	71.4	4754	10.6
LoRA	AIME’24	80.0	14109	7.7
	AIME’25	71.7	16480	7.4
	MATH-500	80.5	4427	10.6
Partial Unfreezing	AIME’24	77.7	13160	7.6
	AIME’25	72.7	14880	7.7
	MATH-500	80.5	4183	10.7

Table 7: Comparison of SFT training strategies on QWQ-32B. Partial unfreezing achieves the best compression while maintaining accuracy.

D LoRA Rank Ablation

To isolate the effect of adapter capacity on supertoken adoption and compression, we sweep LoRA rank $r \in \{1, 2, 4, 8, 16\}$ on Qwen3-30B-A3B (Table 8). Across all ranks, supertoken adoption is consistent ($\sim 10\text{--}13\%$), and accuracy remains robust. Crucially, none of the LoRA configurations produce meaningful compression (token counts remain at or above baseline levels), reinforcing the finding from Section 4.1 that adoption and compression are separate phenomena requiring different degrees of model flexibility.

Method	Benchmark	Acc (%)	Avg. Tok. ($\downarrow$)	Supertoken (%)
LoRA $r=1$	AIME'24	93.3	17695	10.3
	AIME'25	83.3	24120	10.7
	MATH-500	83.0	5832	12.9
LoRA $r=2$	AIME'24	93.3	18589	10.3
	AIME'25	83.3	22756	10.6
	MATH-500	83.6	6024	12.8
LoRA $r=4$	AIME'24	86.7	18517	10.3
	AIME'25	86.7	23729	10.4
	MATH-500	81.8	5920	12.8
LoRA $r=8$	AIME'24	86.7	18170	10.3
	AIME'25	76.7	24371	10.8
	MATH-500	82.8	5814	12.8
LoRA $r=16$	AIME'24	93.3	17655	10.2
	AIME'25	83.4	24055	10.4
	MATH-500	81.4	6024	12.8

Table 8: LoRA rank ablation on Qwen3-30B-A3B. All runs use $\alpha=32$, targets $\{q,k,v,o\}$ -proj, with embed/lm.head as modules_to_save. Supertoken usage is consistent across ranks ($\sim 10\text{--}13\%$); accuracy is robust to rank choice.

E Additional Figures

This section collects figures referenced in the main text that were moved to the appendix for space.

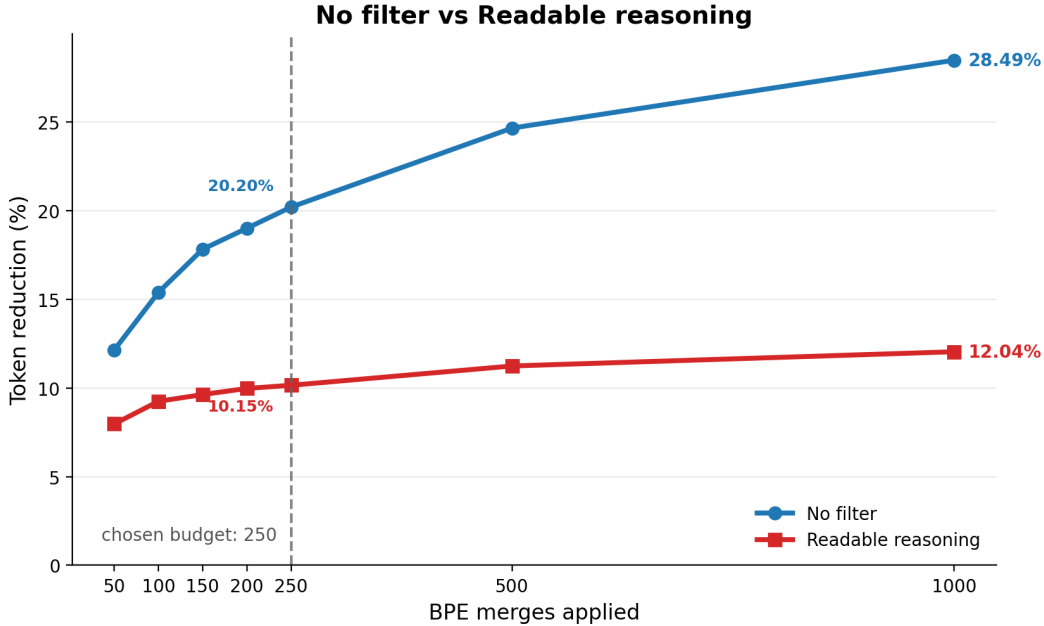


Figure 6: Token compression rate based on merge selection rules and number of merges included.

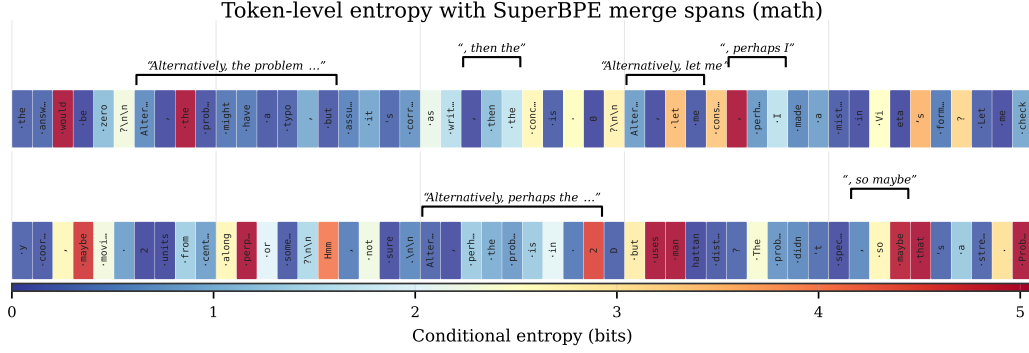


Figure 7: Token-level conditional entropy for two windows from QwQ-32B reasoning traces (blue = low, red = high). Black brackets mark supertoken merge spans (≥ 3 base tokens). Within bracketed spans, continuation tokens are predominantly low-entropy, confirming that structural phrases are near-deterministic once begun (continuation mean: 1.03 bits vs. non-merge mean: 1.27 bits).

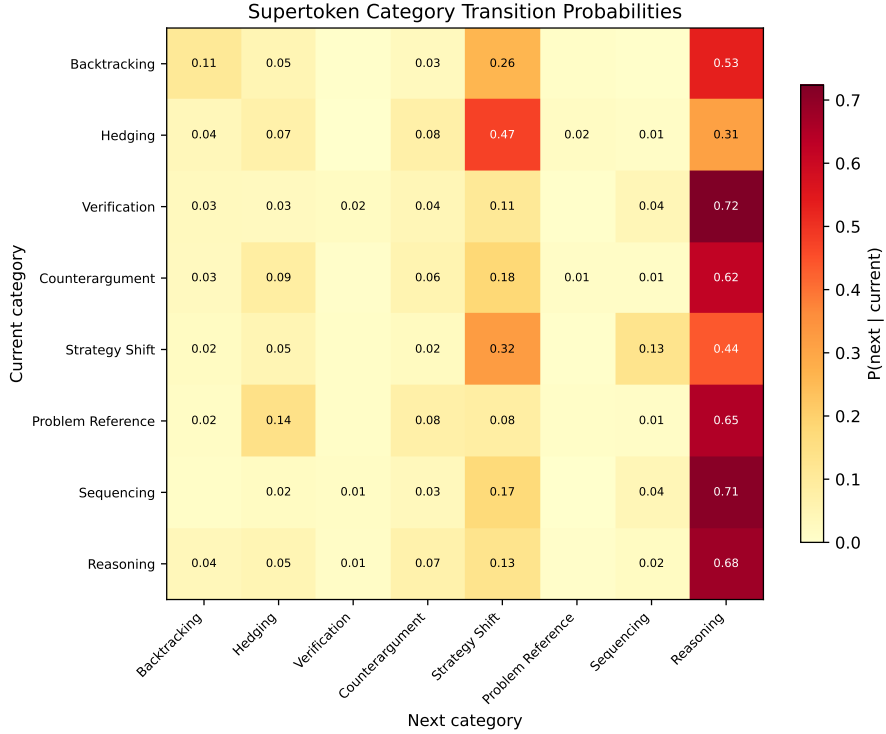


Figure 8: Category transition probabilities aggregated across all evaluation traces. Each cell shows the probability of transitioning from the row category to the column category. High self-transition rates along the diagonal reflect sustained forward computation, while off-diagonal hotspots expose characteristic reasoning patterns such as Backtracking followed by Reasoning (recovery) and Verification followed by Strategy Shift (adaptive re-planning).

F Compression Ceiling Analysis

Applying Equation 2 with $\rho = 15.2\%$ and $h_{\mathcal{M}}/\log |\mathcal{V}| \approx 0.06$, the compression ceiling is $\Delta = 14.3\%$, with $\Delta/\rho = 0.94$. Merged tokens use only 6% of the vocabulary’s information capacity, so nearly all can be absorbed. Continuation tokens contribute $\Delta_{\text{cont}} = 8.0\%$ and merge heads $\Delta_{\text{first}} = 6.2\%$; the effect holds across domains (math 14.9%, code 13.6%, science 11.5%). Figure 9 visualizes this decomposition.

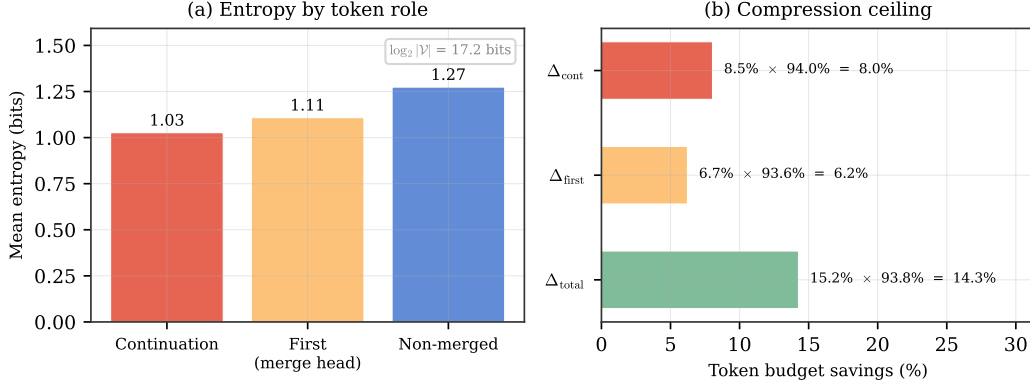


Figure 9: Theoretical compression ceiling. (a) Mean entropy by token role (continuation: 1.03, merge heads: 1.11, non-merged: 1.27 bits), all well below the vocabulary maximum of 17.2 bits. (b) Compression ceiling decomposition ($\Delta = \rho \times (1 - \bar{H} / \log_2 |\mathcal{V}|)$). Compressibility is $\sim 94\%$ across all roles; the combined ceiling $\Delta = 14.3\%$.

G Wall-Clock Latency

Table 9 reports wall-clock latency (mean seconds per sample) for all models and benchmarks. Evaluations are performed using vLLM on $2 \times \text{H200}$ GPUs with greedy decoding. Latency reductions generally track or exceed token count reductions, confirming that the shorter reasoning traces from supertoken adoption translate into practical inference speedups. QwQ-32B shows the most consistent gains (-14.1% average), while DeepSeek-R1-Distill-Llama-70B shows variable results across benchmarks.

Benchmark	Baseline Lat. (s)	SFT Lat. (s)	Δ Lat. (%) ($\downarrow$)
QWQ-32B			
AIME'24	507	466	-8.1
AIME'25	759	649	-14
MATH-500	1538	1252	-18.9
Minerva	620	545	-12
Olympiad	3416	2872	-16
Average	1246	1157	-14.1
QWEN-3-30B-A3B			
AIME'24	2120	1844	-13.1
AIME'25	2453	1922	-21.6
MATH-500	2281	1882	-17.5
Minerva	3595	3571	-0.7
Olympiad	4833	4732	-2.1
Average	3056	2790	-13.1
DeepSeek-R1-Llama-70B-Distill			
AIME'24	618	619.3	+ 0.2
AIME'25	1620	1483	-8.5
MATH-500	2329	1989	-14.6
Minerva	823	831.2	+1.0
Olympiad	9110	8936	-1.9
Average	2900	2772	-4.4

Table 9: Wall-clock latency (mean seconds per sample) for baseline vs. SFT models, served via SGLang or VLLM on H200 GPUs. Latency reductions exceed or track token count reductions from Table 2.

H Cross-Model Entropy Validation

The entropy analysis in Section 4.2 uses QwQ-32B as both generator and scorer, raising the question of whether the observed structural/organic entropy gap is an artifact of a single model’s idiosyncrasies. To address this, we re-score the same QwQ-32B reasoning traces using Qwen3-30B-A3B as the scorer model. Both models share the Qwen tokenizer family, allowing direct token-level comparison.

Table 10 reports the results. Qwen3-30B-A3B assigns higher entropy across all token roles, as expected for a model that did not generate the traces. Crucially, the entropy ordering is preserved: continuation tokens remain the lowest-entropy role (1.36 bits), followed by merge heads (1.47 bits), with non-merged tokens highest (1.98 bits). The structural/organic gap (non-merged – continuation) actually *widens* from 0.26 bits under self-scoring to 0.62 bits under cross-scoring, indicating that the low entropy of structural tokens is not an artifact of the generating model’s own distribution but reflects genuinely predictable patterns in the text. Token-level correlations between the two scorers are high ($r = 0.77$ – 0.84 across roles), confirming that the entropy landscape is consistent across models.

I General Instruction-Following Evaluation

A potential concern with vocabulary expansion and fine-tuning is that it could degrade the model’s general instruction-following ability. To test this, we evaluate all three model

Token role	QwQ-32B (self-scored)	Qwen3-30B-A3B (cross-scored)	Δ	Fraction
Non-merged	1.27	1.98	+0.71	84.9%
Merge head	1.13	1.47	+0.34	7.0%
Continuation	1.01	1.36	+0.35	8.2%
Merged (all)	1.06	1.41	+0.35	15.1%
Gap (non-merged – continuation)	0.26	0.62		

Table 10: Mean conditional entropy (bits) by token role for QwQ-32B reasoning traces (2K samples, 22.9M tokens), scored by the generating model (QwQ-32B, self) vs. a different model (Qwen3-30B-A3B, cross). The structural/organic entropy gap (bottom row) persists and widens under cross-model scoring, confirming the pattern is not an artifact of a single model’s distribution.

families on IFEval (Zhou et al., 2023), a benchmark that measures compliance with explicit formatting and content constraints (e.g., “write exactly 3 paragraphs”, “include the keyword X”). Table 11 reports the strict prompt-level and instruction-level accuracies before and after supertoken SFT.

Model	Baseline		SFT		Δ Prompt (%)
	Prompt (%)	Inst. (%)	Prompt (%)	Inst. (%)	
QwQ-32B	90.5	89.8	84.8	84.2	XXX
Qwen3-30B-A3B	84.8	83.6	XXX	XXX	XXX
DeepSeek-R1-Distill-Llama-70B	XXX	XXX	XXX	XXX	XXX

Table 11: IFEval results before and after supertoken fine-tuning. Prompt-level and instruction-level accuracy remain stable, confirming that vocabulary expansion and SFT do not degrade general instruction-following ability.

J Statistical Significance Analysis

We report 95% confidence intervals (CIs) for all accuracy and completion-token deltas in Table 2 to confirm that accuracy is preserved and that the observed token reductions are real.

Accuracy CIs. Each evaluation run scores N questions (AIME: $N=30$; MATH-500: $N=500$; Minerva: $N=272$; OlympiadBench: $N=674$). The reported accuracy is the mean across 5 independent runs. We compute a conservative 95% CI for the accuracy difference using the standard error of the difference of two independent proportions: $SE(\Delta p) = \sqrt{\frac{p_{\text{base}}(1-p_{\text{base}})}{N} + \frac{p_{\text{sft}}(1-p_{\text{sft}})}{N}}$, with $CI = \Delta p \pm 1.96 \cdot SE(\Delta p)$. This treats each condition as a single evaluation on N questions (ignoring the variance reduction from 5 runs), yielding a conservative upper bound on CI width.

Completion token CIs. Raw per-question token counts have very high variance (problems range from easy to extremely hard), but because each question is answered by both the baseline and SFT model, per-sample pairing cancels out question-level difficulty. The paired-difference standard deviation is typically 20–40% of the raw within-condition standard deviation. We estimate the paired-difference SD as $\sigma_{\text{paired}} \approx 0.24 \times \bar{T}$, where $\bar{T}$ is the mean token count across conditions, reflecting $\sim 30\%$ of a raw SD that is $\sim 80\%$ of the mean. The 95% CI is then $\Delta T \pm 1.96 \cdot \sigma_{\text{paired}} / \sqrt{N}$.

Results. Table 12 reports the estimated CIs. All 15 accuracy CIs include zero, confirming that no accuracy change is statistically significant, including the largest drop (DeepSeek-R1-Distill-Llama-70B on MATH-500, -3.1 pp, CI $[-9.5, +1.1]$). For completion tokens, 9 of 15 reductions are statistically significant (CI excludes zero), concentrated on the larger benchmarks (MATH-500, OlympiadBench) where sample sizes provide sufficient power. On AIME ($N=30$), reductions are directionally consistent but often do not reach significance due to limited sample size, with the exception of the large compressions for DeepSeek-R1-Distill-Llama-70B (-17%) which are significant even at $N=30$.

Model	Benchmark	Accuracy		Completion Tokens		
		ΔAcc (pp)	95% CI	ΔTok	95% CI	Sig.
QwQ-32B	AIME'24	+0.2	$[-20.9, +21.3]$	-922	$[-2092, +248]$	
	AIME'25	+5.5	$[-17.5, +28.5]$	-1771	$[-3125, -417]$	✓
	MATH-500	-0.1	$[-5.0, +4.8]$	-258	$[-349, -167]$	✓
	Minerva	-1.5	$[-9.4, +6.4]$	+94	$[-59, +247]$	
	OlympiadBench	+1.3	$[-4.0, +6.6]$	-328	$[-499, -157]$	✓
Qwen3-30B-A3B	AIME'24	-1.2	$[-15.8, +13.4]$	-1059	$[-2414, +296]$	
	AIME'25	+0.1	$[-18.4, +18.6]$	-974	$[-2606, +658]$	
	MATH-500	+1.2	$[-3.1, +5.5]$	-639	$[-748, -530]$	✓
	Minerva	-1.1	$[-9.2, +7.0]$	-24	$[-126, +78]$	
	OlympiadBench	+2.3	$[-2.8, +7.4]$	-937	$[-1170, -704]$	✓
DeepSeek-R1-Distill-Llama-70B	AIME'24	+5.1	$[-18.4, +28.6]$	-1540	$[-2252, -828]$	✓
	AIME'25	-2.0	$[-27.2, +23.2]$	-1942	$[-2832, -1052]$	✓
	MATH-500	-3.1	$[-9.5, +1.1]$	-160	$[-215, -105]$	✓
	Minerva	-0.6	$[-8.4, +7.2]$	-113	$[-204, -22]$	✓
	OlympiadBench	-2.9	$[-8.2, +2.4]$	-3	$[-118, +112]$	

Table 12: 95% confidence intervals for accuracy and completion-token deltas. **Accuracy:** two-proportion CI treating each condition as a single evaluation on N questions (conservative). All 15 accuracy CIs include zero. **Completion tokens:** paired-difference CI with estimated $\sigma_{\text{paired}} \approx 0.24 \times \bar{T}$. ✓ marks token reductions whose CI excludes zero. Reductions are significant on larger benchmarks; AIME ($N=30$) lacks power except for the large DS-R1-70B compressions.